GHAZI UNIVERSITY D.G KHAN

Assignment

Subject: Programming Fundamentals_Lab2

Professor: Mr. Asad Majeed

Student name: Muhammad Ikram

Class: B.S C.S (1st Semester)

Roll number: 00

Date: 9 Oct 2026

LAB#2

TASK1:

Type and save the following program in visual studio. Run these program and observe their output.

1.#include<iostream>

using namespace std;

int main(){

```
char letter;
letter = 'R';
cout<<letter<<"\n";
letter = 'R';
cout<<letter<<"\n";
cin>>letter;
return 0;
```

}

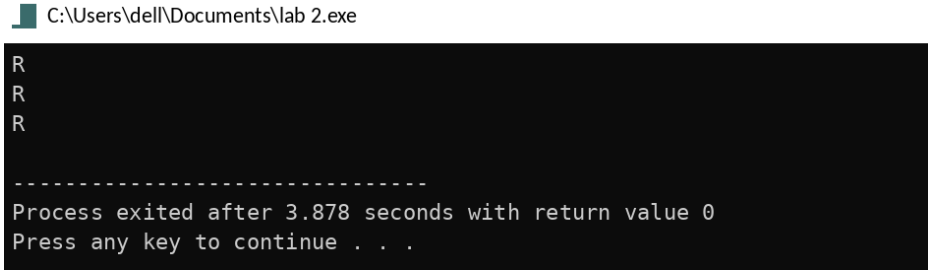
Output:

R

R

R

K



2.#include<iostream>

using namespace std;

int main(){

```
ch  Kletter;
letter = 82;
```

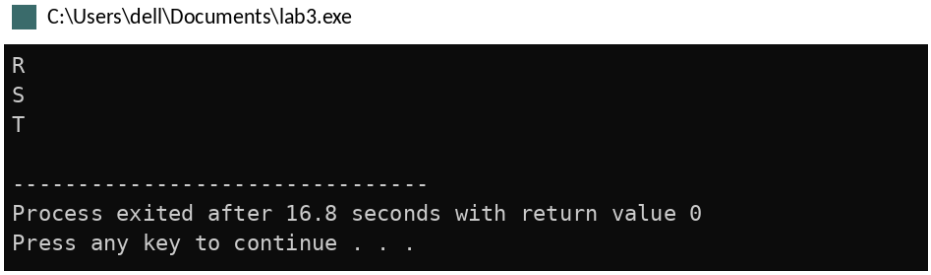
```
    cout<<letter<<'\n';
    letter = 83;
    cout<<letter<<'\n';
    cin>>letter;
    return 0;
}
```

R

S

U

U



3.

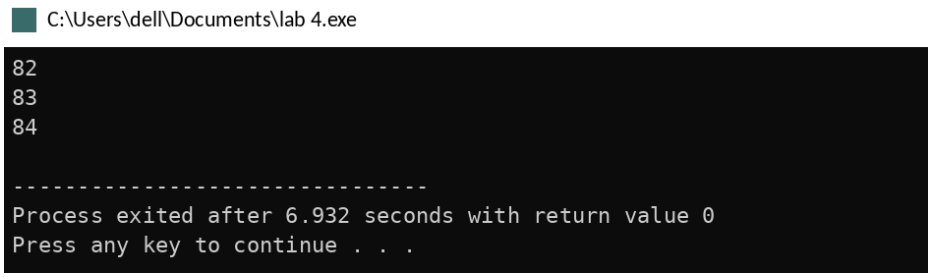
```
#include<iostream>
using namespace std;
int main(){
    int letter;
    letter = 'R';
    cout<<letter<<'\n';
    letter = 'S';
    cout<<letter<<'\n';
    cin>>letter;
    return 0;
}
```

Output:

82

83

90



TASK 2:

Take two integers as input from users apply arithmetic operation on them (+,-,*,/,%) as print them on screen.

```
#include<iostream>
using namespace std;
int main()
{
	int a,b,c,d,e,f,g;
	cout<<"Enter the value of a: ";
	cin>>a;
	cout<<"Enter the value of b: ";
	cin>>b;
	c=a+b;
	cout<<c<<endl;
	d=a-b;
	cout<<d<<endl;
	e=a*b;
	cout<<e<<endl;
	f=a/b;
	cout<<f<<endl;
	g=a%b;
	cout<<g<<endl;
	return 0;
}
```

Output:

Enter the value of a: 34

Enter the value of b: 6

40

28

204

5

4

TASK3:

Write the output of the following code:

```
#include<iostream>
using namespace std;
int main(){
	int val1,val2,val3;
	cout<<"Please enter your 3 numbers: ";
	cin>>val1>>val2>>val3;
	cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
	cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
	system("PAUSE");
	return 0;
}
```

Output:

Please enter your 3 numbers: 123 456 789

123

456

789

789

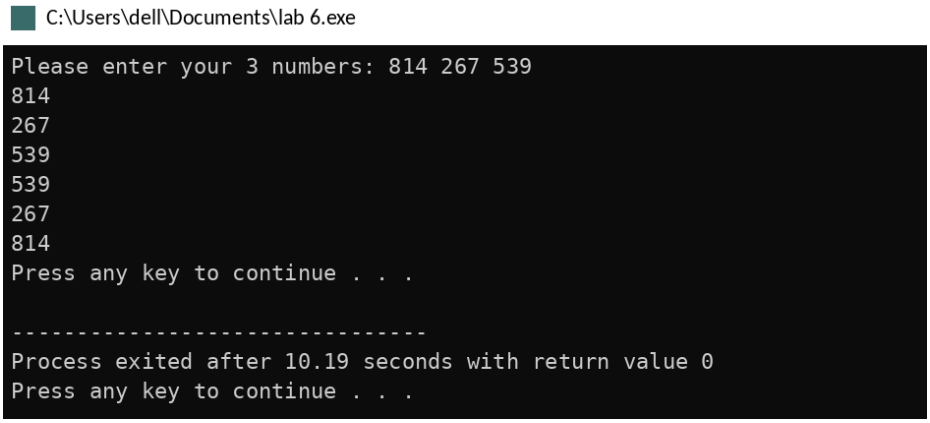
267

539

539

267

814



TASK4:

Write c++ program to calculate the distance between the two points.

NOTE:x1,y1,x2,y2 are all double values.

Formula:

$d=(x_2-x_1)^2+(y_2-y_1)^2$

```
#include<iostream>
using namespace std;
int main(){
	int x1,y1,x2,y2,d;
	cout<<"Enter the value of x1: ";
	cin>>x1;
	cout<<"enter the value of y1: ";
	cin>>y1;
	cout<<"enter the value of x2: ";
	cin>>x2;
	cout<<"enter the value of y2: ";
	cin>>y2;
```

```
    d=(x2-x1)*(x2-x1)+(y2-y1)*(y2-y1);
    cout<<"the distance between two point is: "<<d<<endl;
    return 0;
}
```

Output:

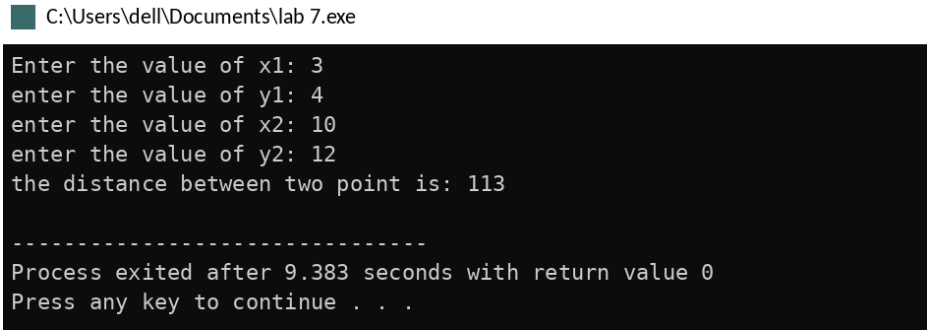
Enter the value of x1: 2

enter the value of y1: 3

enter the value of x2: 8

enter the value of y2: 11

the distance between two point is: 100



HOME TASK1:

Write a program that read in two variable and swaps the value of these variable.

```
#include<iostream>
using namespace std;
int main(){
    int var1,var2,tem;
    cout<<"enter the value of var1: ";
    cin>>var1;
    cout<<"enter the value of var2: ";
    cin>>var2;
    cout<<"\n before swapping"<<endl;
    cout<<"var1 = "<<var1<<endl;
    cout<<"var2 = "<<var2<<endl;
```

```
    tem=var1;
    var1=var2;
    var2=tem;
    cout<<"\n After the swapping: "<<endl;
    cout<<"var1 = "<<var1<<endl;
    cout<<"var2 = "<<var2<<endl;
    return 0;
}
```

Output:

enter the value of var1: 25

enter the value of var2: 7

before swapping

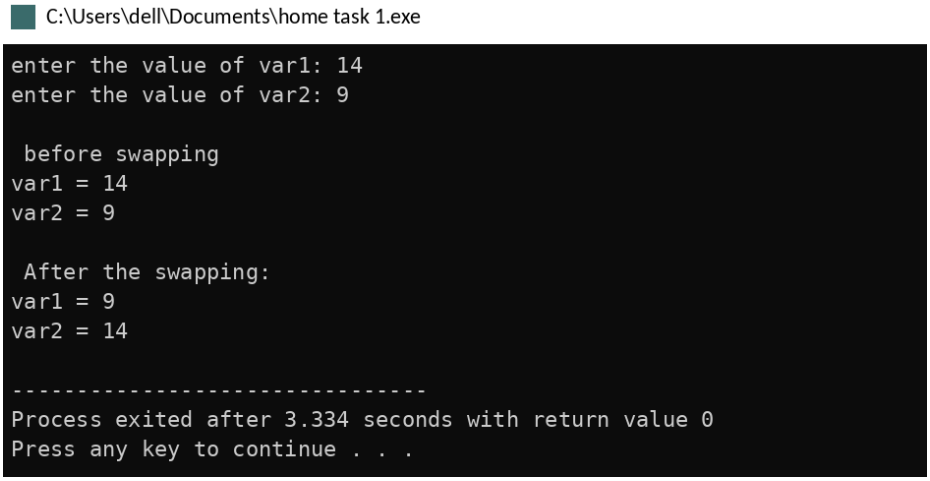
var1 = 25

var2 = 7

After the swapping:

var1 = 7

var2 = 25



HOME TASK2:

Make upper write a program that prompts the user to enter an alphabet in small caps (a,b,c,…,z) and display the entered alphabet into its upper caps (A,B,C,…,Z).

HINT:(ASCII)

```
#include<iostream>

using namespace std;

int main(){

        char letter;

        cout<<"enter a small letter: ";

        cin>>letter;

        letter=letter-32;

        cout<<"uppercase letter= "<<letter<<endl;

        return 0;

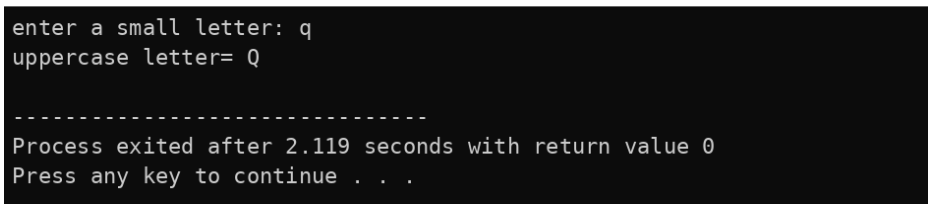
}
```

Output:

enter a small letter: m

uppercase letter= M

ab 8.exe



TASK 6:

Total Purchase: A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $18.25

Price of item 2 = $9.99

Price of item 3 = $14.50

Price of item 4 = $7.75

Price of item 5 = $21.30

Write a program that holds the prices of the five items in five variables. Display each item's price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

```
#include<iostream>
#include<iomanip>
using namespace std;
int main(){
	double cost1=18.25, cost2=9.99, cost3=14.50, cost4=7.75, cost5=21.30;
	double sum, salesTax, grandTotal;
	sum=cost1+cost2+cost3+cost4+cost5;
	salesTax=sum*0.06;
	grandTotal=sum+salesTax;
	cout<<fixed<<setprecision(2);
	cout<<"Item 1 price: $"<<cost1<<endl;
	cout<<"Item 2 price: $"<<cost2<<endl;
	cout<<"Item 3 price: $"<<cost3<<endl;
	cout<<"Item 4 price: $"<<cost4<<endl;
	cout<<"Item 5 price: $"<<cost5<<endl;
	cout<<"Subtotal of sale: $"<<sum<<endl;
	cout<<"Sales tax (6%): $"<<salesTax<<endl;
	cout<<"Total amount: $"<<grandTotal<<endl;
	return 0;
```

}

Output:

Item 1 price: $18.25

Item 2 price: $9.99

Item 3 price: $14.50

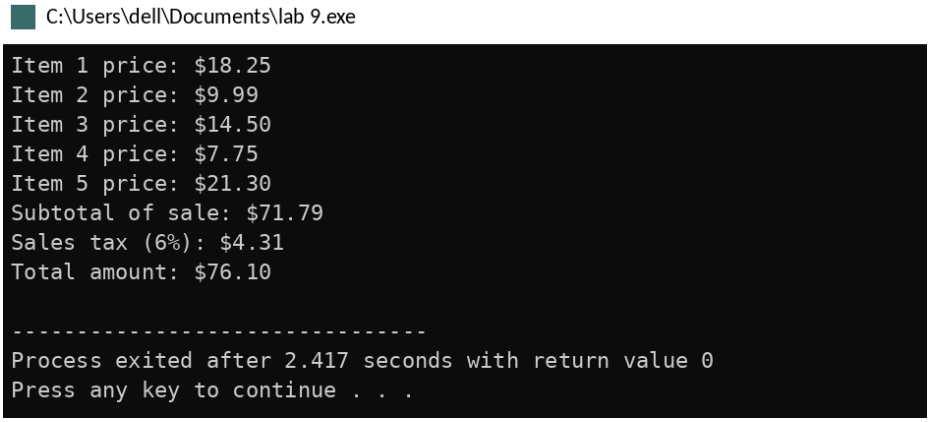
Item 4 price: $7.75

Item 5 price: $21.30

Subtotal of sale: $71.79

Sales tax (6%): $4.31

Total amount: $76.10



BONUS TASK:

Break Number: Write a program that inputs a five-digit integer, separates the integer into its individual digits and prints the digits separated from one another by three spaces each.

```
#include<iostream>
using namespace std;
int main(){
        int number, a, b, c, d, e;
        cout<<"Enter a 5 digit integer: ";
        cin>>number;
        a=number/10000;
        b=(number/1000)%10;
```

```
    c=(number/100)%10;
    d=(number/10)%10;
    e=number%10;
    cout<<a<<"  "<<b<<"  "<<c<<"  "<<d<<"  "<<e<<endl;
    return 0;
}
```

Output:

Enter a 5 digit integer: 57294

5 7 2 9 4

